

HTTP vs. HTTPS:

Securing Your Website with HTTPS Using AWS Certificate Manager: A Step-by-Step Guide

- **HTTP** is not secure, while **HTTPS** ensures security through encryption.
- It's not safe to use HTTP as the data isn't encrypted.
- For HTTPS, you need to upload an SSL/TLS certificate.
- Some companies provide certificates for HTTPS. These certificates encrypt data end-to-end and ensure secure communication.
- Certificates are typically valid for a certain period and can expire, depending on the certificate provider.
- You can also get certificates for your website from providers like DigiCert for secure communication over the internet. After purchasing the certificate, you'll need to attach it to your website.

AWS Certificate Manager (ACM):

- AWS offers **AWS Certificate Manager** (ACM) to manage certificates for AWS resources.
- To set up an HTTPS certificate in AWS, follow these steps:
 1. Log in to the AWS Console.
 2. Go to **Certificate Manager**.
 3. Request a certificate by providing your domain name.
 4. Choose how to validate ownership of the domain, either via **DNS** or **email**. **DNS validation** is recommended, as it's automated and uses RSA encryption keys.
 5. Submit the request.
 6. Accept the validation request by updating DNS records or confirming via email.
- If you're using **Route 53**, AWS will guide you to add a **CNAME** record for validation directly within Route 53, which automates this process.
- Once validated, the certificate will be issued in a few minutes, enabling HTTPS for your domain.

Assigning the Certificate:

1. Attach the certificate to your **Load Balancer** by configuring a listener on port **443** (HTTPS).
2. Configure the listener settings to use the new certificate for secure communication over HTTPS.
3. To ensure all HTTP traffic is redirected to HTTPS, edit the listener for port **80** (HTTP) and configure a redirect to port **443** (HTTPS).